Linear Equations: One, None, or Infinitely Many Solutions

Grade 8 / Pre-Algebra

One solution, no solution, or infinitely many: deciding which, and what it means in context

- Simplify *both* sides first (distribute, then combine like terms).
- **Different x-terms** $\Rightarrow$ the variable survives $\Rightarrow$ **one solution**.
- **Same x-term, different constants** $\Rightarrow$ a false statement such as $7 = 2$ $\Rightarrow$ **no solution**.
- **Same x-term and same constant** $\Rightarrow$ a true statement such as $6 = 6$ $\Rightarrow$ **infinitely many solutions**.

Directions: Show the simplifying on both sides. Then write your answer in one of these three forms: **one solution**, with the value of the variable; **no solution**; or **infinitely many solutions**. Every quantity below is stated with its units.

1. Two saving plans. Ana already has **5** dollars and saves **3** dollars each week. Ben already has **9** dollars and saves **2** dollars each week. After w weeks their amounts are $3w + 5$ and $2w + 9$ dollars. Solve $3w + 5 = 2w + 9$ and classify the equation.

Answer: ________________

2. Two printers. Each printer prints **4** pages per second. Printer A had **7** pages already printed, printer B had **2**. After t seconds the totals are $4t+7$ and $4t+2$ pages.

Seconds t	**1**	**2**	**3**
Printer A: $4t+7$			
Printer B: $4t+2$			

Fill in the table, then solve $4t+7=4t+2$ and classify the equation.

Answer: ___________________

3. Perimeter two ways. A rectangle is x centimetres long and **3** centimetres wide. Sam writes its perimeter as $2(x+3)$; Dana writes it as $2x+6$. Setting the two expressions equal gives $2(x+3)=2x+6$. Expand the left side and classify the equation.

Answer: ___________________

4. Solve if possible and classify: $5(x-2) = 3x+4$

Answer: ____________________

5. Solve if possible and classify: $6x - 3(x+2) = 3x+1$

Answer: ____________________

6. Solve if possible and classify: $-2(4x-6) + 5 = 17 - 8x$

Answer: ____________________

7. **Two gyms.** Gym A charges a **25** dollar joining fee plus **15** dollars each month. Gym B charges a **40** dollar joining fee plus **10** dollars each month. After m months the costs are $25 + 15m$ and $40 + 10m$ dollars. Solve $25 + 15m = 40 + 10m$ and classify the equation.

Answer: ________________

8. The equation $3(2x - 1) = ax + 7$ has **no solution**.

(a) Expand the left side.

Answer: ________________

(b) What value must a have?

Answer: ________________

(c) Explain why that value of a, together with the constant terms, forces the equation to have no solution.

Answer: ________________

9. Classify each equation below *without* fully solving it where you can. Write one solution (with its value), no solution, or infinitely many solutions.

(a) $8x - 3 = 8x + 1$

Answer: ________________

(b) $4(x + 2) = 4x + 8$

Answer: ________________

(c) $5x - 6 = 3x + 10$

Answer: ________________

10. **Two ride prices.** Company P charges **3** dollars plus **2** dollars per mile, so a ride of n miles costs $2n + 3$ dollars. Company Q charges **5** dollars plus **2** dollars per mile, so the same ride costs $2n + 5$ dollars.

(a) Solve $2n + 3 = 2n + 5$ and classify the equation.

Answer: ________________

(b) Explain what your classification means for the two companies' prices: is there *any* ride length for which they charge the same, and why?

Answer: ________________